Yunze Liao

EDUCATION

University of Illinois Urbana-Champaign, Gies College of Business **Champaign, IL**
Master of Science in Financial Engineering (STEM designated program); GPA: 3.8/4.0
Dr.Xiongwei Ju Scholarship Award Winner *Dec* 2024
Concentration: Computer Science & Engineering, Advanced Analytics

University of Illinois Urbana-Champaign **Champaign, IL**
Bachelor of Science in Computer Science + Chemistry, GPA: 3.5/4.0 May 2023

EXPERIENCE

Zelvyn Intelligence LLC **Champaign, IL**
Self-employed *Apr* 2025-Feb 2026

- **Built** a modular C++ event-driven trading engine with explicit separation of market data adapters, signal evaluation, and execution state; implemented WebSocket market connectivity via Boost.Beast with structured JSONL event logging for deterministic replay.
- **Developed** configurable signal framework (ATR, EMA cross, Stoch-RSI) with integrated position sizing and stop-loss / take-profit risk controls.
- **Implemented** real-time telemetry interface (React/Vite) for strategy state and execution event tracking to support iterative testing and validation.

Sinopec Group. **Beijing, China**
Software Engineer *Jan* 2021 – Aug 2021

- **Developed** a backend workflow engine from scratch using Spring Boot, enabling end-to-end processing of forms from initiation to ratification with features such as parallel and ordered countersignatures, automated nodes, and time-off management. Successfully **deployed** the engine across multiple projects, including Sinopec and Sany Heavy Industry.
- **Contributed** to an open-source Instant Messaging platform based on the WebSocket protocol, enhancing secure and efficient communication for China National Machinery Industry Corporation.
- **Participated** in the development of a Test Data Management system focused on IoT data collection using the Modbus protocol. **Designed** workflows for test applications and tasks, and **performed** data mining on test results, with a primary focus on implementing Weibull distribution analysis.

ACTIVITIES

• Low-Code Development Platform **Remote**
Developer *Feb* 2021 – Oct 2021

- **Developed** a platform utilizing object-oriented technology and Domain-Driven Design (DDD) to build and manage data models across multiple domains, defining intricate relationships between them.
- **Implemented** relationship mappings using FreeMarker, enabling the generation of code in Java, C#, PHP, Python, SQL, and other languages, facilitating the compilation of executable application systems.
- **Optimized** the algorithm for building tree structures with HashMap by integrating distributed deployment and multi-threading, increasing data model association relationship generation efficiency by over tenfold.

Financial Report Analyzing LLM – JIA Finance Practicum **Champaign, IL**
Jan 2024 – May 2024

- **Designed and implemented** algorithms to parse and preprocess financial documents (e.g., 10-K, 10-Q), enabling efficient indexing with platforms like LlamaIndex.
- **Conducted comprehensive research** on multiple state-of-the-art large language models (LLMs) and developed a question-and-answer database to quantitatively evaluate their performance.
- **Fine-tuned** large language models to specialize in financial report analysis, enhancing their accuracy and domain-specific effectiveness.

ADDITIONAL INFORMATION

Quant/Data Science: Stochastic Calculus, Optimization, Numerical Analysis, Time-Series Modeling, Derivative Pricing
Technical Skills: Java, C/C++, Python, SQL, R, CUDA, ML (PyTorch/TensorFlow/scikit-learn), NumPy/Pandas
Tools: Docker, Linux, Git, Spring Boot, Flask, FastAPI, Redis, PostgreSQL, MongoDB, Nginx, Maven, LaTeX, Jenkins